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Roku Robot Framework Library

Roku's Robot Framework Library enables keyword-driven testing of apps. The library resides in a Python class that has methods that map directly to keyword names. The keywords take the same arguments as the methods implementing them. The keywords report failures with exceptions, create logs by writing to standard output, and return values using the `return` statement.

Instantiating the library

To create an instance of the Roku Framework Robot Library, provide the following four arguments:

Argument	Description
ip	The IP address of the device to be used for testing.
path	The path to the Roku WebDriver.
timeout	The amount of time (in milliseconds) that commands are allowed to run.
pressDelay	The amount of time (in milliseconds) between keypress commands. This argument works with the Send keys command.

The following example demonstrates how to instantiate the Roku Robot Framework Library:

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```
*** Settings ***
Library ./../Library/RobotLibrary.py  ${ip_address} ${timeout} ${pressDelay}   ${server_path}

*** Variables ***
${ip_address} 127.0.0.1
${server_path} D:/path/to/webDriver/main.exe
${timeout} 20000
${pressDelay} 2000

.py file:
class RobotLibrary:

    def __init__(self, ip, timeout = 0, pressDelay = 0,  path = ""):
        <some code>
```

Keywords

The Roku's Robot Framework Library includes the following keywords:

- Sideload (*available since release 2.0*)
- Launch the app
- Input deep linking data (*available since release 2.0*)
- Get apps
- Send key
- Send keys
- Send word
- Mark timer
- Get timer
- Verify is playback started (*available since release 2.0*)
- Verify is screen loaded (*available since release 2.0*)
- Get child nodes (*available since release 2.1*)
- Get element
- Get elements
- Get focused element
- Verify is app loaded
- Get current app info
- Get device info
- Get player info
- Verify app exists
- Set timeout
- Set press delay
- Get attribute

A keyword will fail if its respective WebDriver endpoint returns a 4xx or 500 error.

Sideload

(*available since release 2.0*)

Keyword	Argument	Description	Example
Sideload	• channel: A zipped package file. • username: Enter rokudev, which is the user name for the Development Application Installer. • password: The password for accessing the Development Application Installer on your Roku device.	Sideloads an app that has been packaged into a zip file. If the Sideload command fails, sideload (/docs/developer-program/getting-started/developer-setup.md#sideloading-channels) the app to be tested and use the Launch the app command.	`Sideload myChannel.zip rokudev y`

Launch the app

Keyword	Argument	Description	Example
Launch the app	• channel_code: The ID of the app to be launched. • contentId: The contentId (/docs/developer-program/discovery/implementing-deep-linking.md#understanding-deep-linking-parameters) of the content to be played. You can include this parameter and the contentType to execute deep linking tests. • mediaType: The mediaType (/docs/developer-program/discovery/implementing-deep-linking.md#understanding-deep-linking-parameters) of the content to be played. You can include this parameter and the contentId to execute deep linking tests.	Launches the app corresponding to the specified channel ID.	Launch the app d

Input deep linking data

(*available since release 2.0*)

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Keyword	Argument	Description	Example
Input deep linking data	• channelId: The ID of the app to be launched. • contentId: The contentId (/docs/developer-program/discovery/implementing-deep-linking.md#understanding-deep-linking-parameters) of the content to be played. You can include this parameter and the contentType to execute deep linking tests. • mediaType: The mediaType (/docs/developer-program/discovery/implementing-deep-linking.md#understanding-deep-linking-parameters) of the content to be played. You can include this parameter and the contentId to execute deep linking tests.	Launches the app corresponding to the specified app ID.	`Input deep linking`

Get apps

Keyword	Description	Example
Get apps	Returns a list of installed apps as an array of objects. Each app object contains the following fields: • title • id • type • version • subtype	`@{apps}=Get Apps`

Send key

Keyword	Arguments	Description	Example
Send key	• key_press: The key to be pressed and released, which may be one of the following: "up", "down", "right", "left", "back, "select", "instantreplay", "play", "stop", "rev", "fwd", and "info". • delay: The delay (in seconds) before the keypresses are executed. This argument is optional, and it defaults to 2 seconds if not specified.	Simulates the press and release of the specified key.	`Send key up 2`

Send keys

Keyword	Arguments	Description	Example
Send keys	• sequence: An array containing the sequence of keys to be pressed and released (for example, down, down, down, down, select). • delay: The delay (in seconds) before the keypresses are executed. This argument is optional, and it defaults to 2 seconds if not specified.	Simulates the sequence of keypresses and releases.	**Variables*** @{keys}= down down down down select ***Test cases*** Send keys ${keys} 1

Send word

Keyword	Arguments	Description	Example
Send word	• word: The specified word to be entered. • delay: The delay (in seconds) before the entry of each letter in the specified word. This argument is optional, and it defaults to 2 seconds if not specified.	Simulates the press and release of each letter in a word.	`Send word Hello`

Mark timer

(*available since release 2.0*)

Keyword	Description	Example
Mark timer	Starts the timer.	`Mark timer`

Get timer

(*available since release 2.0*)

Keyword	Description	Example
Get timer	Returns the number of milliseconds elapsed since the timer was last started.	`${time} = Get timer`

Verify is playback started

Keyword	Arguments	Description	Example
Verify is playback started	• retries: The number of requests that can be made before returning false. This argument is optional, and it defaults to 10 if not specified. • delay: The delay (in seconds) between retries. This argument is optional, and it defaults to 1 second if not specified.	Verify playback has started on the Roku media player. This keyword fails if player state is not "play".	`Verify is playback started 10 1`

Verify is screen loaded

Keyword	Arguments	Description	Example
Verify is screen loaded	• data: An object with locators for elementData and parentData (parentData is optional). See the WebDriver element command (/docs/developer-program/dev-tools/automated-channel-testing/web-driver.md#POST-v1/session/:sessionId/elements) command for more information. • retries: The number of requests that can be made before returning false. This argument is optional, and it defaults to 10 if not specified. • delay: The delay (in seconds) between retries. This argument is optional, and it defaults to 1 second if not specified.	Verify that the screen is loaded based on the provided element data.	***Variables*** &{ElementData}= us @{ElementArray}= &{ElementParams} *** Test Cases *** Verify is screen loade

Get child nodes

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Keyword	Arguments	Description	Example
Get child nodes	• parentNode: The parent node for which the child nodes are to be retrieved.	Retrieves the child component of the specified node.	***Variables*** &{LabelData}= using=text value=Live Gaming &{IndexData}= using=attr attribute=inc @{LabelArray}= &{LabelData} &{Index @{ParamArray}= &{PosterData} ***Test Cases*** &{focusedEl}= get focusedElement @{Nodes}= Get child nodes ${focusedEl} ${ParamArray}

- locator: An array containing search criteria for the child nodes to be retrieved. The locator has the following syntax:

```
using=attribute, tag, or text  attribute=specific attribute  value=tag or attribute value
```

Get element

Keyword	Arguments	Description	Example
Get element	• data: An object with locators for elementData and parentData (parentData is optional). See the WebDriver element command (/docs/developer-program/dev-tools/automated-channel-testing/web-driver.md#POST-v1/session/:sessionId/elements) for more information. • delay: The delay (in seconds) between retries. This argument is optional, and it defaults to 1 second if not specified.	Searches for an element on the page based on the specified locator starting from the screen root. Returns information on the first matching element.	***Variables*** &{ElementData}= u @{ElementArray}= &{ElementParams} ***Test Cases*** &{element}= Get eler

Get elements

Keyword	Arguments	Description	Example
Get elements	• data: An object with locators for elementData and parentData (parentData is optional). See the WebDriver element command (/docs/developer-program/dev-tools/automated-channel-testing/web-driver.md#POST-v1/session/:sessionId/elements) for more information. • delay: The delay (in seconds) between retries. This argument is optional, and it defaults to 1 second if not specified.	Searches for elements on the page based on the specified locators starting from the screen root. Returns information on the matching elements.	***Variables*** &{ElementData}= u @{ElementArray}= &{ElementParams} ***Test Cases*** @{elements}= Get

Get focused element

Keyword	Description	Example
Get focused element	Return the element on the screen that currently has focus. See the WebDriver active element command (/docs/developer-program/dev-tools/automated-channel-testing/web-driver.md#get-v1/session/:sessionId/element/active) for more information.	`&{element}= Get focused element`

Verify is channel loaded

Keyword	Arguments	Description	Example
Verify is channel loaded	• id: The ID of the app to be launched. Use `dev` to verify a sideloaded app. • retries: The number of requests that can be made before returning false. This argument is optional, and it defaults to 10 if not specified. • delay: The delay (in seconds) between retries. This argument is optional, and it defaults to 1 second if not specified.	Verify that the specified app has been launched. This keyword fails if the provided app ID does not match a valid channel.	`Verify is channel loaded dev`

Get current channel info

Keyword	Description	Example
Get current channel info	Returns an object containing information about the app currently loaded. This object has the following fields:	&{channel}=Get current char

Key	Type	Description
sessionId	string	The advertising ID of the device
status	int	A status code summarizing the result of the command.
value	array	
value[i].Title	string	The title of the app.
value[i].ID	string	The ID of the app.
value[i].Version	string	The build version of the app.
value[i].Subtype	string	"ndka"/"rsga"
value[i].Type	string	"menu"/"appl"

Get device info

Keyword	Description	Example
Get device info	Returns an object containing the information about the device. This object has the following fields:	&{info}=Get device info

Key	Type	Description
sessionId	string	The advertisement ID of the device.
status	int	A status code summarizing the result of the command.
value	object	
value.vendorName	string	The vendor of the device.
value.modelName	string	The model of the device.
value.language	string	The language of the device.
value.country	string	The country of the device.
value.ip	string	The IP address of the device.
value.timeout	int	The specified timeout for WebDriver client requests.
value.pressDelay	int	The specified delay between key presses.

Get player info

Keyword	Description	Examp
Get player info	Returns an object containing information about the Roku media player. This object has the following fields:	&{play

Key	Type	Description
sessionId	string	The advertising ID of the device
status	int	A status code summarizing the result of the command
value	object	
value.error	string	Indicates whether there was a playback error. If no error occurred, this is set to "false"
value.state	string	Indicates the current playback state ("play", "pause", "resume", and so on)
value.format	object	The format element contains the following attributes: *audio*, *caption*, *container*, *drm*, *video*, and *res*.
value.format.audio	string	The audio compression method ("aac", "aac_adts", and so on.)
value.format.caption	string	The closed caption format ("608_708", for example). This value is set to "none" if there are no captions.

Keyword	Description	Example

Key	Type	Description
value.format.container	string	The container format ("hls", for example)
value.format.drm	string	The encoding type. If no encoding is used, this us set to "none".
value.format.video	string	The format of the currently playing video stream ("mpeg4-15", for example)
value.format.res	string	The resolution of the currently playing video stream ("1280X720", for example).
value.buffering	object	The buffering element contains the following attributes: *current*, *max*, *target*.
value.buffering.current	string	The current buffering speed (in kbps).
value.buffering.max	string	The maximum possible buffering speed (in kbps).
value.buffering.target	string	The target buffering speed (in kbps).
value.newStream	object	The newStream element contains the following attribute: *speed*.

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Keyword	Description	Examp

Key	Type	Description
value.newStream.speed	string	The current playback speed (in bps)
value.position	string	The time of the current position in the stream, expressed as the elapsed time (in ms) since the start of stream or UTC time, depending on the content.
value.duration	string	The duration of the video being played (in seconds). This becomes valid when playback begins and may change if the video is dynamic content, such as a live event.
value.isLive	string	A flag indicating whether the video being played is a live stream.
value.runtime	string	The runtime of the video being played (in seconds).

Keyword	Description	Examp

Key	Type	Description
value.streamSegment	object	The streamSegment attribute contains Information about the video segment that is currently streaming. This is only meaningful for segmented video transports, such as DASH and HLS This element contains the following attributes: *bitrate*, *mediaSequence*, *segmentType*, and *time*.
value.streamSegment.bitrate	string	The bitrate of the video segment (in bps).
value.streamSegment.mediaSequence	string	The HLS media sequence ID of the segment in the video.
value.streamSegment.segmentType	string	The type of data in the segment, which may be one of the following values: "audio", "video", "captions", "mux".
value.streamSegment.time	string	The chunk start time.

Verify is channel exist

Keyword	Arguments	Description	Example
Verify is channel exist	• apps: An array containing currently installed on the device. • id: The ID of the app to be verified. Use `dev` to verify a sideloaded app.	Verifies the specified app is installed on the device. This keyword fails if the apps array does not contain the app specified in the id argument.	`@{apps}= Get apps` `Verify is channel exist @{apps} dev`

Set timeout

Keyword	Arguments	Description	Example
Set timeout	• timeout: The amount of time (in milliseconds) that Web driver client requests are allowed to run.	Sets the timeout for Web driver client requests.	`Set timeout 5000`

Set press delay

Keyword	Arguments	Description	Example
Set press delay	• delay: The interval (in milliseconds) to be used between key presses.	Sets the delay between key presses. This keyword works with the Send keys keyword.	`Set press delay 2000`

Keyword	Arguments	Description	Example
Get attribute			
Get attribute	• element: An object that contains element information (attributes, child nodes). • attr: The name of the attribute to be retrieved.	Get attribute value. This keyword fails if an element does not contain the specified attribute.	***Variables*** &{ElementData}= using=text value=sc @{ElementArray}= &{ElementData} &{ElementParams} elementData=${Ele ***Test Cases*** &{element}= Get element ${ElementPar ${attrValue}= Get attribute ${element

Sample test cases

The Roku automated app testing repository (https://github.com/rokudev/automated-channel-testing) includes a set of sample Robot Framework test cases that can be executed on their corresponding SceneGraph Developer Extensions (SGDEX) sample apps (https://github.com/rokudev/SceneGraphDeveloperExtensions/tree/master/samples). For example, you can execute the SGDEX GridView test case (test_3_Grid.robot), which will sideload the corresponding sample app (3_Grid) on your device, and then view the test output. You can reference these samples when developing test scripts for the automated testing of your development apps.

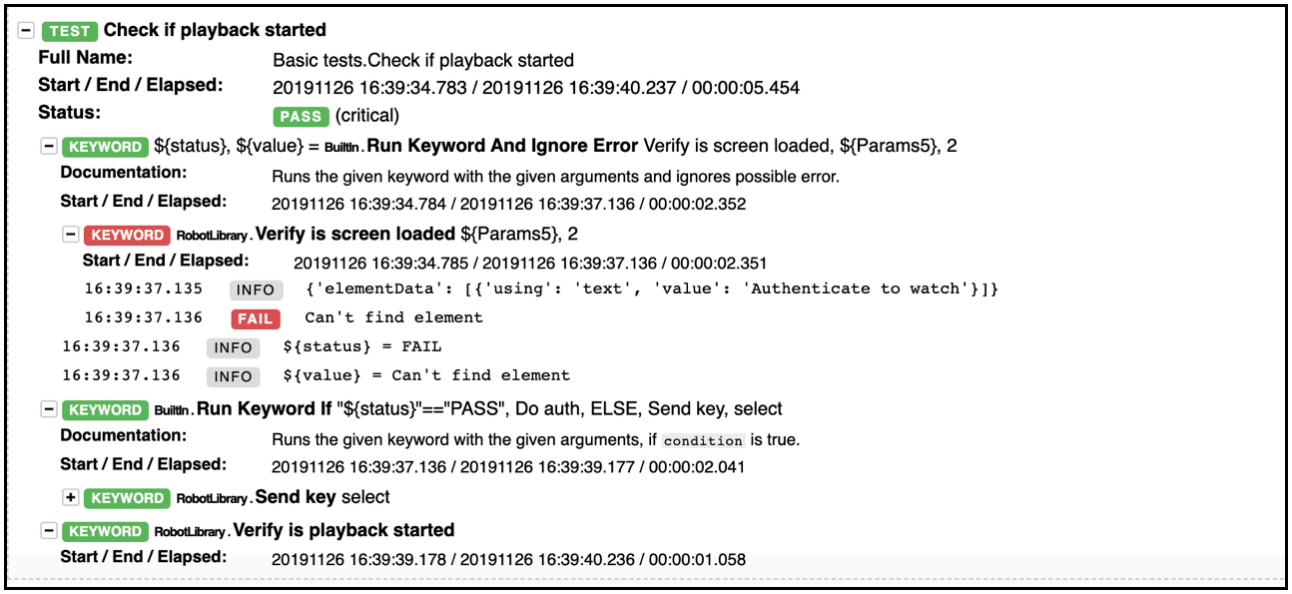
> Before running a sample test case, you need to update the sideload command in the test case with the Roku device password.

The Basic_tests.robot sample (https://github.com/rokudev/automated-channel-testing/blob/master/RobotLibrary/Tests/Basic_tests.robot) demonstrates how to create a simple test case that checks whether a user is authenticated before playing content using the Roku Robot Framework Library:

```
*** Settings ***
Documentation  Basic smoke tests
Variables  ./../Library/variables.py
Library  ../../Library/RokuLibrary.py  ${ip_address}  ${timeout}  ${pressDelay}  ${server_path}
Library  Collections

*** Variables ***
${channel_code}  dev
&{DATA2}=  using=text  value=Barack Gates, Bill Obama
@{DATA2Array}=  &{DATA2}
&{Params2}=  elementData=${DATA2Array}
&{DATA3}=  using=text  value=Please enter your username
@{DATA3Array}=  &{DATA3}
&{Params3}=  elementData=${DATA3Array}
&{DATA4}=  using=text  value=Please enter your password
@{DATA4Array}=  &{DATA4}
&{Params4}=  elementData=${DATA4Array}
@{KEYS}=   down  down  down  down  select
&{DATA5}=  using=text  value=Authenticate to watch
@{DATA5Array}=  &{DATA5}
&{Params5}=  elementData=${DATA5Array}

*** Test Cases ***
Channel should be launched
   Side load  ../sample/channel.zip   rokudev   aaaa
   Verify is channel loaded    ${channel_code}

Check if details screen showed
   Send key  select  4
   Verify is screen loaded    ${Params2}

Check if playback started
   ${status}  ${value}=  Run Keyword And Ignore Error  Verify is screen loaded  ${Params5}  2
   Run keyword if   "${status}"=="PASS"  Do auth
   ...  ELSE  Send key  select
   Verify is playback started  20  2

*** Keywords ***
Do auth
   Send key  select
   Verify is screen loaded   ${Params3}
   Send word  user
   Send keys  ${KEYS}
   Verify is screen loaded   ${Params4}
   Send word  pass
   Send keys  ${KEYS}
```

Viewing the test case report and log

After you run a test case that uses the Roku Robot Framework Library, you can view the generated report and log files in the specified output directory. The report summarizes the test case and provides statistics on the percentage of individual tests that passed/failed. The log details the success/failure of the individual keywords used in each test case.



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